

Fire Safety Equipment Inspection Register

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1. The Problem

- Fire safety equipment carries paper tags showing the last inspection.
- Nobody maintains a central list of all equipment.
- The Consequence: Expired units are often discovered during an audit, or far worse, when one is needed during an emergency and fails.

2. The Solution

- A centralized, digital register that tracks every unit's location.
- Automatically calculates next due dates.
- Instantly highlights which units are overdue or approaching expiry.
- Simple, responsive, and easy to use on mobile devices.

3. Main Screen (List & Search)

- Clean, responsive list showing critical details at a glance.
- Real-time search by ID, Type, or Location.
- Filter by status (Valid, Expiring Soon, Expired, Unknown).
- The total record count updates instantly as filters are applied.

4. Detail View & Derived Figure

- Opens instantly upon tapping a record.
- Derived Figure: A large, color-coded banner showing exact days until expiry (or days overdue) is placed at the very top.
- The safety officer sees the most important number first, avoiding the need to read through a long table.

5. Calculation Logic

- 1. Parse 'next_due' string into a Date object.
- 2. Fetch current local date and strip out the time of day.
- 3. Calculate difference in milliseconds.
- 4. Convert to days by dividing by $(1000 * 60 * 60 * 24)$ and round up.
- 5. If negative, flag equipment as 'Overdue'.

6. Edge Cases & Mobile UI

- Loading State: Simulated network delay with a spinner.
- Empty Search: Clear message when no results match criteria.
- Awkward Cases: Missing locations show 'Unknown Loc', missing equipment shows a prominent yellow warning banner.
- Mobile First: Minimum touch targets of 44px height, stacked flexbox layout.

7. Current Status & Future Improvements

- What works: Search, filter, calculations, detail view, and mobile UI.
- What is unfinished / Next improvement: Data is currently static JSON.
- Next step: Implement a backend or LocalStorage solution to allow adding new records directly from the UI.